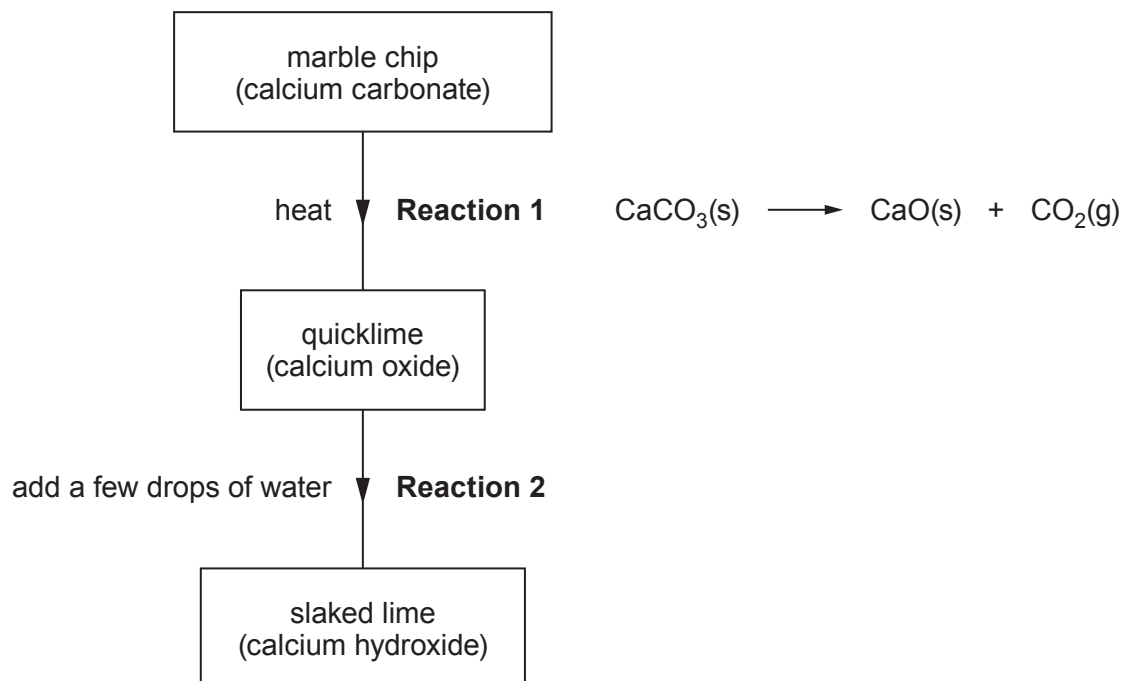


4. (a) The flow diagram shows the reactions necessary to make slaked lime (calcium hydroxide) from a marble chip (calcium carbonate).



- (i) In **Reaction 1**, one compound breaks down when heated to form two products. Complete the name of this type of reaction. [1]
- thermal
- (ii) Give **two** observations you would expect to make as **Reaction 2** happens. [2]
- 1
- 2
- (iii) Complete the symbol equation for the reaction between calcium oxide and water in **Reaction 2**. [1]



- (b) (i) The relative formula mass of calcium hydroxide is 74.

Calculate the number of moles of calcium hydroxide in 2.96 g.

[2]

Number of moles = mol

- (ii) Explain why calcium hydroxide is used to treat some soils.

[2]

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